

PHYSICS 550
Homework #6
Due on Nov 3, 2023

1. G 4.6 (10 points)
2. G 4.9 (10 points)
3. G 4.13(10 points)
4. (5 points) Derive Eq. (4.13) of Griffith
5. (10 points) You have a mass M in a potential which is $V(x)=\infty$ for $x<0$ and $V(x)=Fx$ for $x>0$ and $F>0$. At $t=0$, the wavefunction is $\Psi(x,0)=Axe^{(-x/L)}$ where A and L are positive constants. Compute the expectation value of the energy.